

0 5

Two enzymes, **P** and **Q**, are proteins with quaternary structure which catalyse the same reaction, but they have different amino acid sequences.

0 5 . 1

Define the **quaternary structure** of a protein.

[1 mark]

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0 5 . 2

Explain how two enzymes with different amino acid sequences can catalyse the same reaction.

[2 marks]

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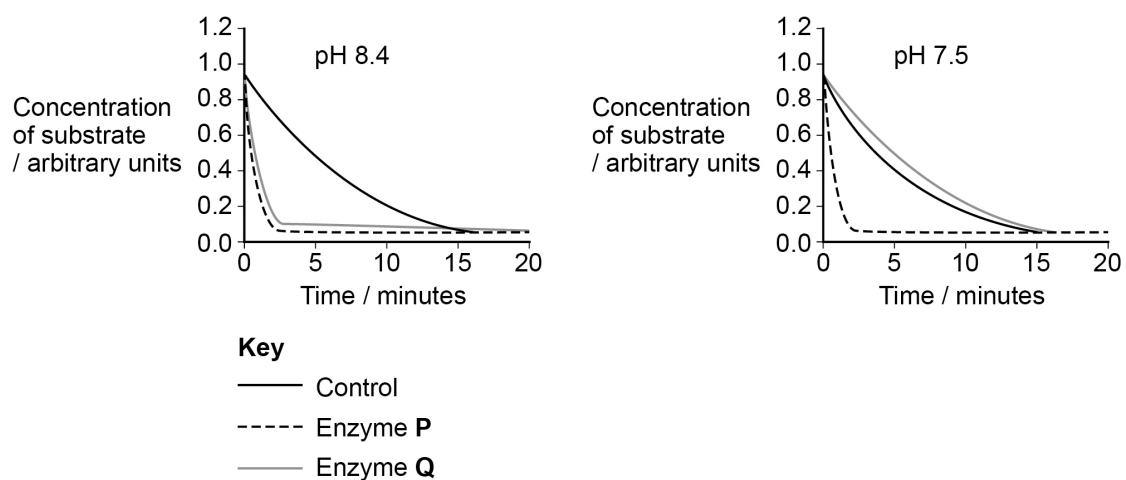
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Scientists investigated the effect of pH 8.4 and pH 7.5 on the activity of enzymes **P** and **Q**.

**Figure 8** shows their results.

**Figure 8**



0 5 . 3

Describe what the scientists should place in the control tubes in this investigation.

**[3 marks]**

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0 5 . 4

Give **three** conclusions you can make from **Figure 8**.**[3 marks]**

1 

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2 

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3 

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